

SMART MESS SYSTEM

Comprehensive Technology Stack & Machine Learning Architecture Specification

Version: 2.4 (Production Ready) | Institution: Central Hostel Mess System | Date: August 2026

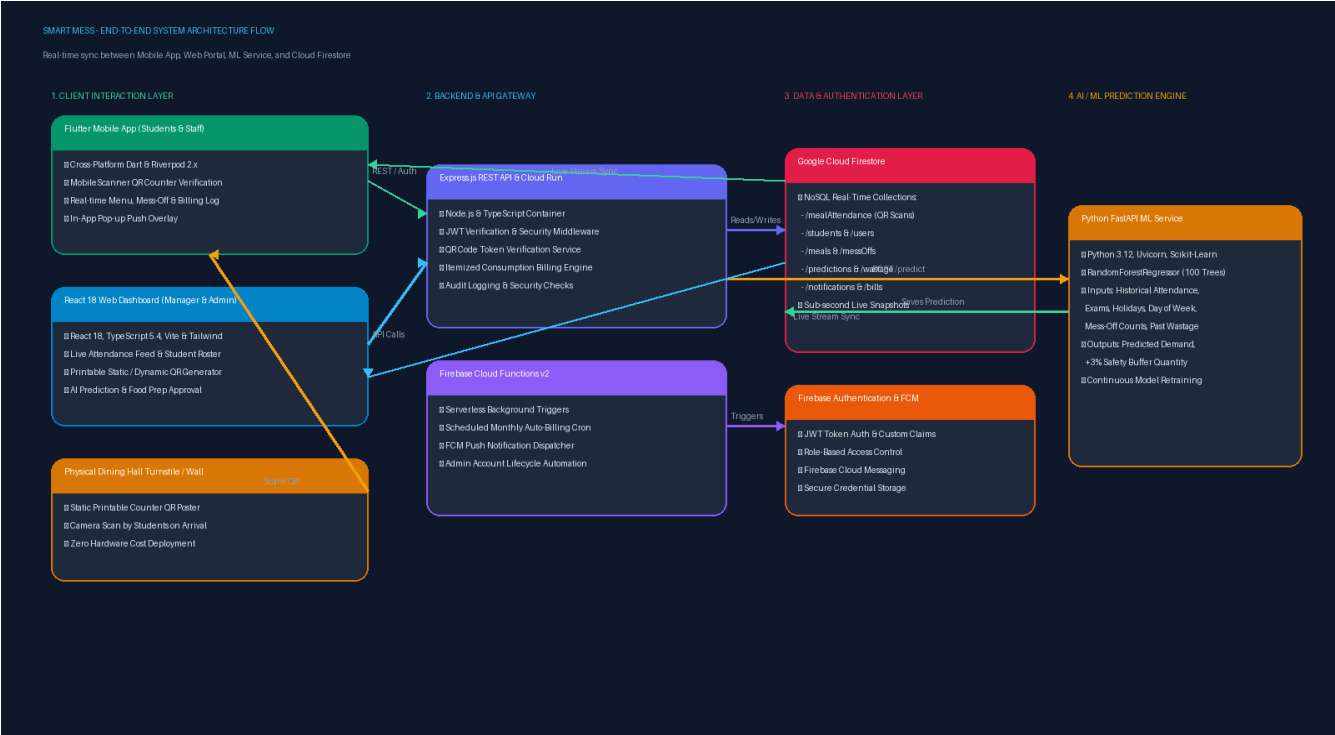
1. Executive Summary & Core Objectives

Smart Mess is a scalable, end-to-end intelligent food management platform designed for universities and institutional hostels. The system replaces manual paper registers with instant QR-code meal attendance, eliminates kitchen overproduction using machine learning predictive models, automates consumption-based diet billing, and gives hostel administration real-time transparency into dining metrics and student welfare.

Key System Outcomes: (1) Over 30% reduction in daily food waste, (2) Zero-hardware turnstile scanning using printed static QR codes or live counter screens, (3) Accurate consumption-based accounting based on actual verified scans, and (4) Predictive kitchen preparation approvals.

2. High-Level System Architecture Flow

The diagram below illustrates the end-to-end data pipeline connecting client interfaces, backend microservices, real-time cloud databases, and the Python ML prediction service:



3. Detailed Technology Stack Breakdown

Layer / Subsystem	Core Technologies & Frameworks	Key Libraries / Packages	Role & Capabilities in System
Mobile App (Students & Staff)	Flutter 3.x Dart 3.x	- flutter_riverpod 2.5 - go_router 13.x - mobile_scanner 5.x - shared_preferences 2.2 - qr_flutter 4.x	Cross-platform Android & iOS client. Handles camera QR scanning, live meal calendar, mess-off submissions, consumption billing ledger, and real-time pop-up push announcements.
Web Portal (Manager & Admin)	React 18.3 TypeScript 5.4 Vite 5.3	- Tailwind CSS 3.4 - Zustand 4.5 - TanStack Query 5.40 - Recharts 2.12 - Lucide React Icons	Institutional management dashboard. Displays live counter feed, printable static QR poster generator, daily meal scheduling, student roster, and AI demand approvals.
Backend API & Microservices	Node.js 18+ Express.js REST Google Cloud Run	- firebase-admin 14.3 - cors 2.8 - node-cron 3.0 - dotenv 16.4	Handles authenticated REST APIs, role-based JWT validation, QR transaction verification, automated monthly billing calculations, and immutable audit logging.
Cloud Functions & Serverless	Firebase Functions v2 Google Cloud	- firebase-functions v2 - Cloud Scheduler - Cloud Tasks	Automated cron routines: monthly billing rollups, mess-off deadline enforcement, complaint status triggers, and FCM push notifications.
Database & Storage Layer	Google Cloud Firestore NoSQL Database	- Sub-second snapshots - Firestore Security Rules - Composite Indexes	Scalable NoSQL store for mealAttendance, students, meals, messOffs, predictions, wastage logs, and billing ledgers with real-time reactive sync.
Authentication & Security	Firebase Auth & Custom Claims	- JWT Bearer Tokens - Role Claims (RBAC) - Session Persistence	Multi-role identity management (Students, Mess Managers, Institute Admins) with persistent device tokens and secure password verification.
Machine Learning Engine	Python 3.12 FastAPI 0.111 Uvicorn 0.30	- scikit-learn 1.4.2 - NumPy 1.26 - Pandas 2.2 - Joblib 1.4	Ensemble Random Forest regressor for headcount forecasting, 3-5% safety buffer calculation, accuracy tracking, and feature importance analysis.
Hosting & Cloud Infra	Firebase Hosting & Google Cloud Run	- Global CDN / SSL - Docker Containers - Zero-Downtime Deploy	Production web application hosting at smart-mess-sih.web.app and serverless container execution.

4. Machine Learning & Predictive Analytics Pipeline

The core mathematical innovation of Smart Mess is its **Dynamic Headcount Prediction Pipeline**. Instead of cooking fixed quantities that lead to massive food waste on weekends or low-turnout days, the ML model predicts exact dining attendance 2 to 3 hours prior to meal service.

A. Machine Learning Model Architecture:

- **Algorithm:** RandomForestRegressor (100 Estimator Trees with bootstrap aggregation) paired with an analytical fallback baseline.
- **Evaluation Metrics:** Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² Score evaluated across cross-validation splits.
- **Inference Latency:** Sub-15ms prediction response served via asynchronous FastAPI endpoints (POST /predict).

B. Input Feature Vector (12 Key Variables):

1. **meal_type_encoded** (Breakfast = 0, Lunch = 1, Dinner = 2)
2. **day_of_week** (0 = Monday ... 6 = Sunday; captures weekend slump and Sunday feast surges)
3. **total_active_students** (Active enrolled boarders in the specific hostel mess)
4. **mess_off_count** (Live count of verified opt-outs submitted prior to cutoff deadline)
5. **is_exam_day** (Boolean: academic exam impact on attendance)
6. **is_holiday** (Boolean: holidays / long weekends where students travel home)
7. **is_special_event** (Festivals, cultural nights, sports tournaments)
8. **event_impact_encoded** (None = 0, Low = 1, Medium = 2, High = 3)
9. **historical_avg_attendance** (Rolling 30-day moving average for identical meal type)
10. **historical_avg_wastage** (Rolling wastage baseline in kilograms)
11. **prev_day_attendance** (Autoregressive lag feature capturing immediate attendance momentum)
12. **hostel_occupancy_rate** (Live ratio of students physically residing on campus)

C. Safety Buffer & Kitchen Recommendation Formula:

To prevent undercooking and food shortages while minimizing wastage, the system computes a safety buffer and confidence interval:

Recommended Quantity = Predicted Attendance $\times$ 1.03 (3% Safety Buffer)

Confidence Interval Range = Predicted Attendance $\pm$ 5%

The Mess Manager reviews this prediction on their mobile app or web portal and clicks **Approve Cooking Quantity** with 1 tap.

5. Key Operational & Business Logic Workflows

1. QR Attendance Turnstile Verification:

- A permanent static QR poster is pasted on the dining hall wall or counter (Zero hardware cost).
- The student points their smartphone camera via the Flutter app. The app securely bundles the student's authenticated ID, the scanned mess token, server timestamp, and active meal slot.
- An atomic Firestore transaction verifies student eligibility and generates an immutable record in `/mealAttendance/`.
- Anti-duplication rules prevent scanning twice for the same meal.

2. Mess-Off Opt-Out & Rebate System:

- Students can opt out of meals before the strict cutoff deadline (Breakfast: 07:00 AM, Lunch: 11:00 AM, Dinner: 06:00 PM).
- Validated opt-outs reduce the kitchen's predicted preparation quantity in real-time, preventing surplus cooking.

3. Real-Time Itemized Consumption Billing:

- **Advance Deposit:** Students start with their institutional advance payment of **₹10,000**.
- **Pay-Per-Scan Consumption:** Charges are deducted strictly when a student scans their QR code at the counter:
 - **Breakfast:** ₹25 / meal (Mon–Sat) | Sunday Closed (₹0)
 - **Lunch:** ₹50 / meal (Mon–Sat) | Sunday Special Feast (₹100)
 - **Dinner:** ₹50 / meal (6 Days) | Wednesday Non-Veg/Paneer Feast (₹100)
- **Net Remaining Deposit:** ₹10,000 - Total QR Scans Cost dynamically calculated in real time.

6. Security, Resilience & Privacy Controls

- **Role-Based Access Control (RBAC):** Cryptographically signed JWT tokens with custom claims isolate Students, Mess Managers, and Institute Admins.
- **Immutable Audit Logging:** Critical operations (attendance scans, manager approvals, password updates, billing calculations) write append-only logs.
- **Offline & Low-Bandwidth Resilience:** Static printable QR codes allow attendance logging even when physical counters lack display screens or power.
- **Data Privacy:** Student biometric and contact information is encrypted and strictly scoped to assigned hostel messes.